CHAPTER 03

Bonding

Independent exercise solving

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

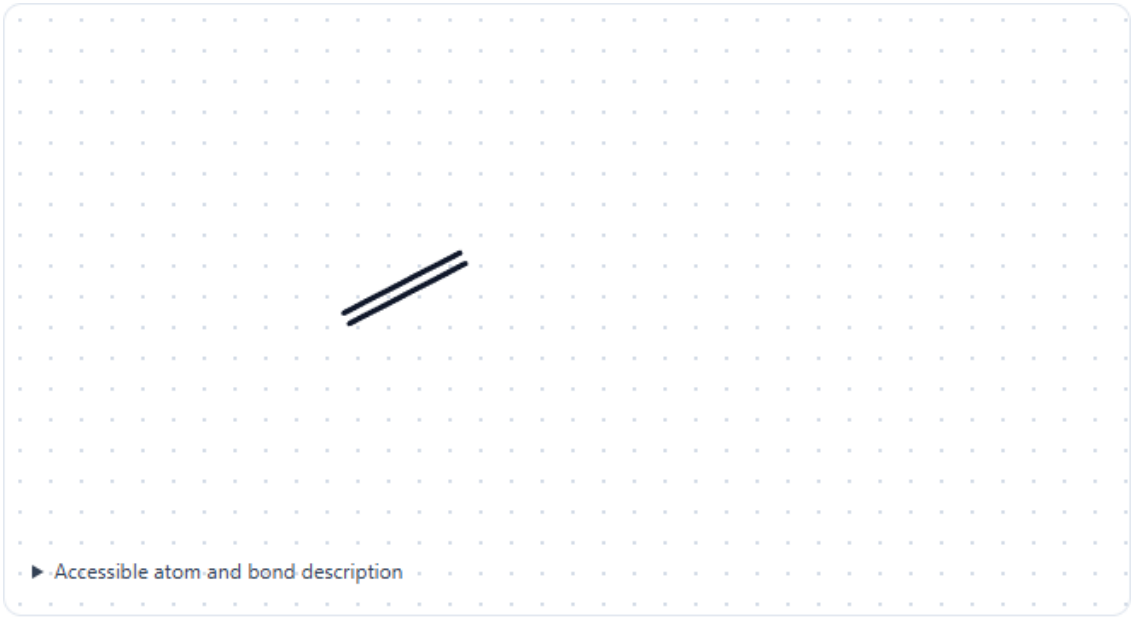
Source snapshot SHA-256: 8a59db9f1bd641519bbc53e66a48dc91e2f98451fcadaed885ac48986c9e616b

BLIND PASS: answer keys, author molecule names/formulas, hints, explanations and answer overlays are withheld. Solve every numbered stage independently. Report the stage ID, answer, derivation, and uncertainty. Question wording and displayed diagram captions are retained because they are visible task data.

Do not open the full review packet in this chat before saving your independent attempt. The full packet contains keys and lessons that would contaminate a blind attempt.

quantum | Independent exercise

Screenshot: quantum-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2]]

quantum:stage-1 | For a 2p electron, which n and l pair is allowed?

Answer type: choice

Option 0 (A): n=2, l=1

Option 1 (B): n=2, l=2

Option 2 (C): n=1, l=1

quantum:stage-2 | How many orbitals are in a p subshell?

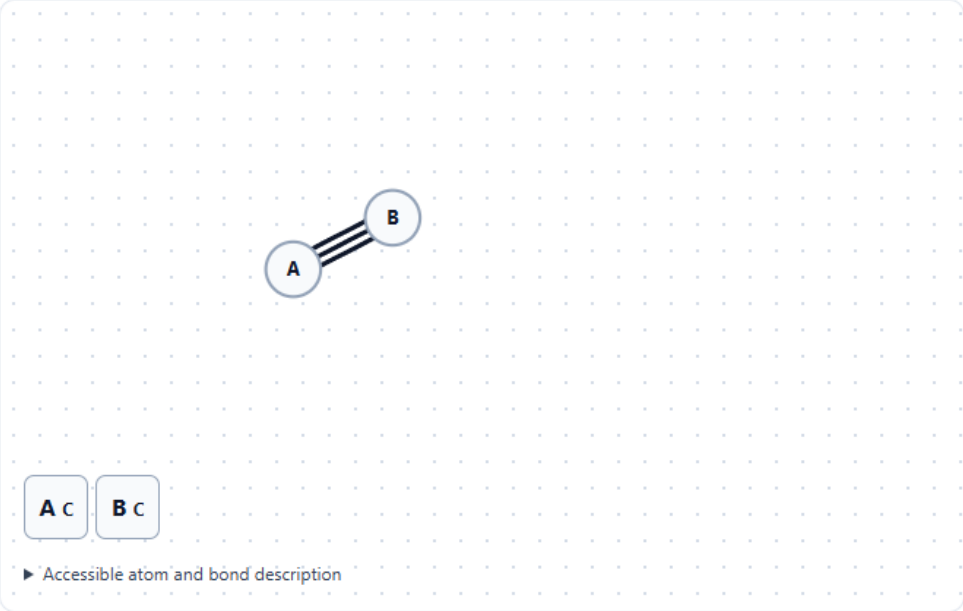
Answer type: number | unit: orbitals

quantum:stage-3 | How many electrons can a filled p subshell hold?

Answer type: number | unit: electrons

bond-count | Independent exercise

Screenshot: bond-count-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 3]]

bond-count:stage-1 | Select the atoms joined by the triple bond.

Answer type: atoms

bond-count:stage-2 | How many pi bonds are present?

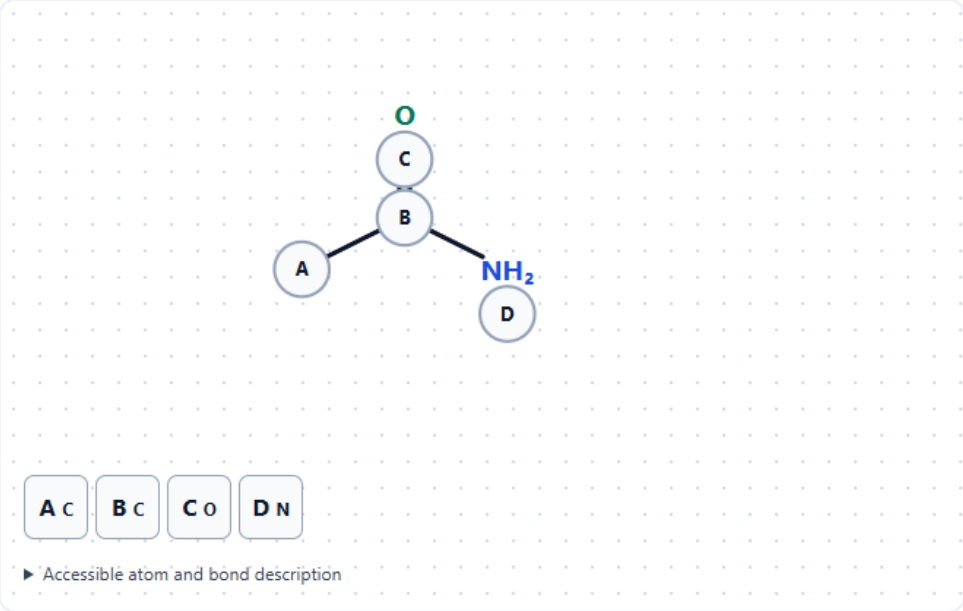
Answer type: number | unit: pi bonds

bond-count:stage-3 | Including implicit C–H bonds, how many sigma bonds are present?

Answer type: number | unit: sigma bonds

amide-resonance | Independent exercise

Screenshot: amide-resonance-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=55.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=115.00, y=25.00; charge=0; explicit H override=none

3 (D): N; x=175.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1]]

amide-resonance:stage-1 | Select the amide nitrogen.

Answer type: atoms

amide-resonance:stage-2 | Why is the amide group approximately planar?

Answer type: choice

Option 0 (A): Nitrogen lone-pair conjugation with C=O

Option 1 (B): All nitrogen atoms are planar

Option 2 (C): C–N is a freely rotating pure single bond

amide-resonance:stage-3 | For acetate, resonance contributors differ in…

Answer type: choice

Option 0 (A): Positions of atoms

Option 1 (B): Electron placement only

Option 2 (C): Molecular formula